A First Course in Abstract Algebra

Part 7: Advanced Group Theory

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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34 Isomorphism Theorems

Exercise 34.1: A homomorphism from $\mathbb{Z}_{12}$ to $\mathbb{Z}_3$

Plan. An additive homomorphism is determined by the image of 1, so $\phi(x) = 2x \pmod 3$. **(a)** $2x \equiv 0 \pmod 3$ holds exactly when $3 \mid x$. Thus $K = \{0, 3, 6, 9\}$. **(b)** The distinct cosets are

$$K = \{0, 3, 6, 9\}, \quad 1 + K = \{1, 4, 7, 10\}, \quad 2 + K = \{2, 5, 8, 11\}.$$

(c) The first-isomorphism map is $x + K \mapsto \phi(x)$, hence $K \mapsto 0$, $1 + K \mapsto 2$, $2 + K \mapsto 1$. Changing x by an element of K does not change $\phi(x)$, so the map is well defined; it is a bijective homomorphism, not merely a matching of three sets.

Exercise 34.2: A homomorphism from $\mathbb{Z}_{18}$ to $\mathbb{Z}_{12}$

(a) Here $\phi(x) = 10x \pmod{12}$. Its kernel solves $12 \mid 10x$, equivalently $6 \mid x$, so $K = \{0, 6, 12\}$. **(b)** The six cosets are $r + K = \{r, r + 6, r + 12\}$ for $0 \le r \le 5$:

$$\{0, 6, 12\},\ \{1, 7, 13\},\ \{2, 8, 14\},\ \{3, 9, 15\},\ \{4, 10, 16\},\ \{5, 11, 17\}.$$

(c) Multiples of 10 modulo 12 give $\phi[\mathbb{Z}_{18}] = \{0, 2, 4, 6, 8, 10\}$. **(d)** Under $r + K \mapsto 10r \pmod{12}$, the cosets listed above map respectively to $0, 10, 8, 6, 4, 2$. Addition is preserved because $10(r + s) = 10r + 10s$; the kernel calculation makes the quotient map injective.

Exercise 34.3: The second isomorphism theorem in $\mathbb{Z}_{24}$

Plan. In an additive cyclic group, $H + N = \langle \gcd(4, 6) \rangle$ and $H \cap N = \langle \operatorname{lcm}(4, 6) \rangle$. **(a)** Thus $H + N = \{0, 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22\}$ and $H \cap N = \{0, 12\}$. **(b)** The cosets in $(H + N)/N$ are

$0 + N$	$\{0, 6, 12, 18\}$
$2 + N$	$\{2, 8, 14, 20\}$
$4 + N$	$\{4, 10, 16, 22\}$

(c) Put $I = H \cap N$. The cosets in H/I are

$0 + I$	$\{0, 12\}$
$4 + I$	$\{4, 16\}$
$8 + I$	$\{8, 20\}$

(d) Use the representatives from H: $h + N \mapsto h + I$. Explicitly the correspondences are

$$0 + N \longmapsto 0 + I, \quad 4 + N \longmapsto 4 + I, \quad 2 + N \longmapsto 8 + I.$$

The representative from H matters: an arbitrary displayed integer representing a coset modulo N need not belong to H. If $h_1 + N = h_2 + N$ with $h_i \in H$, then $h_1 - h_2 \in H \cap N = I$, proving well-definedness. The same observation proves injectivity, and every $h + I$ is reached.

Exercise 34.4: The second isomorphism theorem in $\mathbb{Z}_{36}$

Plan. In an additive cyclic group, $H + N = \langle \gcd(6, 9) \rangle$ and $H \cap N = \langle \operatorname{lcm}(6, 9) \rangle$. **(a)** Thus $H + N = \{0, 3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33\}$ and $H \cap N = \{0, 18\}$. **(b)** The cosets in $(H + N)/N$ are

$0 + N$	$\{0, 9, 18, 27\}$
$3 + N$	$\{3, 12, 21, 30\}$
$6 + N$	$\{6, 15, 24, 33\}$

(c) Put $I = H \cap N$. The cosets in H/I are

$0 + I$	$\{0, 18\}$
$6 + I$	$\{6, 24\}$
$12 + I$	$\{12, 30\}$

(d) Use the representatives from H: $h + N \mapsto h + I$. Explicitly the correspondences are

$$0 + N \longmapsto 0 + I, \quad 6 + N \longmapsto 6 + I, \quad 3 + N \longmapsto 12 + I.$$

The representative from H matters: an arbitrary displayed integer representing a coset modulo N need not belong to H. If $h_1 + N = h_2 + N$ with $h_i \in H$, then $h_1 - h_2 \in H \cap N = I$, proving well-definedness. The same observation proves injectivity, and every $h + I$ is reached.

Exercise 34.5: The third isomorphism theorem in $\mathbb{Z}_{24}$

(a) $H = \langle 4 \rangle$, so the following table lists every coset of H in G:

coset	elements
$0 + \langle 4 \rangle$	$\{0, 4, 8, 12, 16, 20\}$
$1 + \langle 4 \rangle$	$\{1, 5, 9, 13, 17, 21\}$
$2 + \langle 4 \rangle$	$\{2, 6, 10, 14, 18, 22\}$
$3 + \langle 4 \rangle$	$\{3, 7, 11, 15, 19, 23\}$

(b) $K = \langle 8 \rangle$. Its cosets in G are

coset	elements
$0 + \langle 8 \rangle$	$\{0, 8, 16\}$
$1 + \langle 8 \rangle$	$\{1, 9, 17\}$
$2 + \langle 8 \rangle$	$\{2, 10, 18\}$
$3 + \langle 8 \rangle$	$\{3, 11, 19\}$
$4 + \langle 8 \rangle$	$\{4, 12, 20\}$
$5 + \langle 8 \rangle$	$\{5, 13, 21\}$
$6 + \langle 8 \rangle$	$\{6, 14, 22\}$
$7 + \langle 8 \rangle$	$\{7, 15, 23\}$

(c) In H/K there are two elements, namely K and $4 + K$; their underlying sets are

$$K = \{0, 8, 16\}, \qquad 4 + K = \{4, 12, 20\}.$$

(d) Write $C_r = r + K$ for the cosets in part (b). The cosets of H/K in G/K are precisely

$$\{C_r, C_{r+4}\} \qquad (0 \le r < 4).$$

This displays the elements of each outer coset: each is itself one of the two explicitly listed K-cosets.
(e) The required isomorphism is

$$r + H \longmapsto \{C_r, C_{r+4}\} = (r + K) + (H/K).$$

Adding representatives preserves addition of outer cosets. Its kernel before quotienting is H, which explains both well-definedness and injectivity.

Exercise 34.6: The third isomorphism theorem in $\mathbb{Z}_{36}$

(a) $H = \langle 9 \rangle$, so the following table lists every coset of H in G:

coset	elements
$0 + \langle 9 \rangle$	$\{0, 9, 18, 27\}$
$1 + \langle 9 \rangle$	$\{1, 10, 19, 28\}$
$2 + \langle 9 \rangle$	$\{2, 11, 20, 29\}$
$3 + \langle 9 \rangle$	$\{3, 12, 21, 30\}$
$4 + \langle 9 \rangle$	$\{4, 13, 22, 31\}$
$5 + \langle 9 \rangle$	$\{5, 14, 23, 32\}$
$6 + \langle 9 \rangle$	$\{6, 15, 24, 33\}$
$7 + \langle 9 \rangle$	$\{7, 16, 25, 34\}$
$8 + \langle 9 \rangle$	$\{8, 17, 26, 35\}$

(b) $K = \langle 18 \rangle$. Its cosets in G are

coset	elements
$0 + \langle 18 \rangle$	$\{0, 18\}$
$1 + \langle 18 \rangle$	$\{1, 19\}$
$2 + \langle 18 \rangle$	$\{2, 20\}$
$3 + \langle 18 \rangle$	$\{3, 21\}$
$4 + \langle 18 \rangle$	$\{4, 22\}$
$5 + \langle 18 \rangle$	$\{5, 23\}$
$6 + \langle 18 \rangle$	$\{6, 24\}$
$7 + \langle 18 \rangle$	$\{7, 25\}$
$8 + \langle 18 \rangle$	$\{8, 26\}$
$9 + \langle 18 \rangle$	$\{9, 27\}$
$10 + \langle 18 \rangle$	$\{10, 28\}$
$11 + \langle 18 \rangle$	$\{11, 29\}$
$12 + \langle 18 \rangle$	$\{12, 30\}$
$13 + \langle 18 \rangle$	$\{13, 31\}$
$14 + \langle 18 \rangle$	$\{14, 32\}$
$15 + \langle 18 \rangle$	$\{15, 33\}$
$16 + \langle 18 \rangle$	$\{16, 34\}$
$17 + \langle 18 \rangle$	$\{17, 35\}$

(c) In H/K there are two elements, namely K and $9 + K$; their underlying sets are

$$K = \{0, 18\}, \qquad 9 + K = \{9, 27\}.$$

(d) Write $C_r = r + K$ for the cosets in part (b). The cosets of H/K in G/K are precisely

$$\{C_r, C_{r+9}\} \qquad (0 \le r < 9).$$

This displays the elements of each outer coset: each is itself one of the two explicitly listed K-cosets. **(e)** The required isomorphism is

$$r + H \longmapsto \{C_r, C_{r+9}\} = (r + K) + (H/K).$$

Adding representatives preserves addition of outer cosets. Its kernel before quotienting is H, which explains both well-definedness and injectivity.

Exercise 34.7: Normality of an intersection

Let $h \in H$ and $x \in H \cap N$. Since H is a subgroup, $hxh^{-1} \in H$. Since $N \trianglelefteq G$ and $h \in G$, also $hxh^{-1} \in N$. Thus $h(H \cap N)h^{-1} \subseteq H \cap N$. Apply the same argument to h^{-1} to obtain the reverse inclusion. Hence $H \cap N \trianglelefteq H$, directly by the conjugation definition of normality.

Exercise 34.8: Successive quotient groups

(a) For $gH \in A$ and $kH \in B$, $(gH)(kH)(gH)^{-1} = (gkg^{-1})H \in B$ because $K \trianglelefteq G$. This proves $B \trianglelefteq A$; replacing K by L proves $C \trianglelefteq A$. Also $K \subseteq L$ implies $B \subseteq C$. If the original inclusion is strict, so is this one: $l \in L \setminus K$ gives $lH \in C \setminus B$. **(b)** The third isomorphism theorem gives $A/B \cong G/K$ and identifies C/B with L/K. Therefore

$$(A/B)/(C/B) \cong (G/K)/(L/K) \cong G/L.$$

Explicitly the final map sends the class represented successively by gH, then by $(gH)B$, to gL. Its kernel consists precisely of representatives in L.

Exercise 34.9: Two complementary normal subgroups

Because $K, L \trianglelefteq G$, the product KL is a subgroup containing both; hence $KL = K \vee L = G$. Restrict the quotient map $G \to G/K$ to L. Every $g = kl$ has $gK = lK$ (normality permits changing kl to lk'), so $l \mapsto lK$ is onto. Its kernel is $L \cap K = \{e\}$, so it is an isomorphism $L \cong G/K$. Interchanging K and L gives $K \cong G/L$. These conclusions require normality, not merely trivial intersection.

35 Series of Groups

Throughout this section $D_4 = \langle r, s\rangle$ denotes the order-8 square-symmetry group, with $r^4 = s^2 = e$ and $srs = r^{-1}$. We follow the textbook's definitions of principal series and solvability; qualifications are stated where they affect an answer.

Exercise 35.1: Refining the $10\mathbb{Z}$ and $25\mathbb{Z}$ series

Insert subgroups to obtain

$$\{0\} < 50\mathbb{Z} < 10\mathbb{Z} < 5\mathbb{Z} < \mathbb{Z},$$
$$\{0\} < 50\mathbb{Z} < 25\mathbb{Z} < 5\mathbb{Z} < \mathbb{Z}.$$

The first factor in each is $50\mathbb{Z} \cong \mathbb{Z}$. The remaining factors are, respectively, $\mathbb{Z}_5, \mathbb{Z}_2, \mathbb{Z}_5$ and $\mathbb{Z}_2, \mathbb{Z}_5, \mathbb{Z}_5$. Thus the factors match after reordering. Here $a\mathbb{Z}/b\mathbb{Z} \cong \mathbb{Z}_{b/a}$ when $a \mid b$; remember that a larger integer labels a smaller subgroup.

Exercise 35.2: Refining the $60\mathbb{Z}$ and $245\mathbb{Z}$ series

A common initial nonzero subgroup is $2940\mathbb{Z}$, since $\operatorname{lcm}(60, 245) = 2940$. Use

$$\begin{aligned}\{0\} < 2940\mathbb{Z} < 420\mathbb{Z} < 60\mathbb{Z} < 20\mathbb{Z}\\ < 10\mathbb{Z} < 5\mathbb{Z} < \mathbb{Z},\\ \{0\} < 2940\mathbb{Z} < 1470\mathbb{Z} < 735\mathbb{Z} < 245\mathbb{Z}\\ < 49\mathbb{Z} < 7\mathbb{Z} < \mathbb{Z}.\end{aligned}$$

Every original subgroup remains. In the first series the factors after $2940\mathbb{Z} \cong \mathbb{Z}$ have orders $7, 7, 3, 2, 2, 5$; in the second they have orders $2, 2, 3, 5, 7, 7$. Each factor is cyclic, so this is an isomorphism of series.

Exercise 35.3: Refinements in $\mathbb{Z}_{24}$

All subgroups here are cyclic and normal. The refinements

$$\{0\} < \langle 12\rangle < \langle 6\rangle < \langle 3\rangle < \mathbb{Z}_{24},$$
$$\{0\} < \langle 8\rangle < \langle 4\rangle < \langle 2\rangle < \mathbb{Z}_{24}$$

have factor orders $2, 2, 2, 3$ and $3, 2, 2, 2$. They retain respectively $\langle 3\rangle$ and $\langle 8\rangle$. Each factor is cyclic of the indicated prime order, giving the required matching.

Exercise 35.4: Refinements in $\mathbb{Z}_{72}$

Use

$$\{0\} < \langle 36\rangle < \langle 18\rangle < \langle 9\rangle < \langle 3\rangle < \mathbb{Z}_{72},$$
$$\{0\} < \langle 24\rangle < \langle 12\rangle < \langle 6\rangle < \langle 3\rangle < \mathbb{Z}_{72}.$$

The orders of these subgroups are respectively $1, 2, 4, 8, 24, 72$ and $1, 3, 6, 12, 24, 72$. Consecutive order ratios give cyclic factors of orders $2, 2, 2, 3, 3$ and $3, 2, 2, 2, 3$. These agree up to order and all original terms remain.

Exercise 35.5: Refining two series of $\mathbb{Z}^2$

Put $L = 60\mathbb{Z} \times 80\mathbb{Z}$ and $0 = \{(0,0)\}$. The refinements are

$$\begin{aligned} 0 < L < 60\mathbb{Z} \times 20\mathbb{Z} < 60\mathbb{Z} \times \mathbb{Z} \\ < 10\mathbb{Z} \times \mathbb{Z} < \mathbb{Z}^2, \\ 0 < L < 10\mathbb{Z} \times 80\mathbb{Z} < \mathbb{Z} \times 80\mathbb{Z} \\ < \mathbb{Z} \times 20\mathbb{Z} < \mathbb{Z}^2. \end{aligned}$$

The initial factor is $L \cong \mathbb{Z}^2$ in both. The other factors are $\mathbb{Z}_4, \mathbb{Z}_{20}, \mathbb{Z}_6, \mathbb{Z}_{10}$ in the first and $\mathbb{Z}_6, \mathbb{Z}_{10}, \mathbb{Z}_4, \mathbb{Z}_{20}$ in the second. This follows by quotienting the coordinate that changes at each step; the unchanged coordinate cancels.

Exercise 35.6: All composition series of $\mathbb{Z}_{60}$

Plan. A cyclic group has exactly one subgroup U_d of each order $d \mid 60$, namely $U_d = \langle 60/d \rangle$. A cyclic factor is simple exactly when its order is prime. Consequently *all* the series are

$$U_1 < U_{p_1} < U_{p_1 p_2} < U_{p_1 p_2 p_3} < U_{60},$$

where (p_1, p_2, p_3, p_4) runs over the distinct permutations of $(2, 2, 3, 5)$. This is an explicit parametrization with no duplication: the subgroup orders recover the sequence of primes. There are $4!/2! = 12$ such series. All have factors $\mathbb{Z}_2, \mathbb{Z}_2, \mathbb{Z}_3, \mathbb{Z}_5$, in some order, so all are isomorphic.

Exercise 35.7: All composition series of $\mathbb{Z}_{48}$

Let $U_d = \langle 48/d \rangle$ be the unique subgroup of order d. The possible successive subgroup orders are

$$\begin{gathered} 1, 3, 6, 12, 24, 48 \\ 1, 2, 6, 12, 24, 48 \\ 1, 2, 4, 12, 24, 48 \\ 1, 2, 4, 8, 24, 48 \\ 1, 2, 4, 8, 16, 48. \end{gathered}$$

Replacing every order d by U_d lists all five series. Indeed each step must multiply the order by a prime, and $48 = 2^4 \cdot 3$, so the only choice is the position of the factor 3 among five steps. Each series has four factors $\mathbb{Z}_2$ and one $\mathbb{Z}_3$.

Exercise 35.8: Composition series of $\mathbb{Z}_5^2$

Regard $\mathbb{Z}_5^2$ as a two-dimensional vector space over $\mathbb{Z}_5$. Its order-5 subgroups are exactly the six lines

$$L_t = \langle (1, t) \rangle \ (t = 0, 1, 2, 3, 4), \qquad L_\infty = \langle (0, 1) \rangle.$$

They exhaust the possibilities: a nonzero vector with nonzero first coordinate scales uniquely to $(1, t)$; otherwise it scales to $(0, 1)$. The six composition series are $0 < L_t < \mathbb{Z}_5^2$, including $t = \infty$. Both factors have order 5. No other composition series is possible, since simple factors of an abelian group have prime order.

Exercise 35.9: Composition series of $S_3 \times \mathbb{Z}_2$

Put $T = A_3 \times \{0\}$, $B = S_3 \times \{0\}$, $A = A_3 \times \mathbb{Z}_2$, $U = \{e\} \times \mathbb{Z}_2$, and $C = \{(\sigma, \epsilon) : \operatorname{sgn}(\sigma) = (-1)^\epsilon\}$. Then all four series are

$$1 < T < A < G, \qquad 1 < U < A < G,$$
$$1 < T < B < G, \qquad 1 < T < C < G.$$

Here $A \cong \mathbb{Z}_6$, while $B \cong C \cong S_3$. To check exhaustion, the commutator subgroup of G is T, and $G/T \cong \mathbb{Z}_2^2$. A maximal normal subgroup must therefore be one of the three inverse images of its lines, namely A, B, C (a simple quotient of this solvable group has prime order). The cyclic group A has the two indicated series, and each S_3 factor has only its A_3 series. The factor orders are $3, 2, 2$ or $2, 3, 2$, so all are isomorphic.

Exercise 35.10: Composition series of $\mathbb{Z}_2 \times \mathbb{Z}_5 \times \mathbb{Z}_7$

The product is cyclic of order 70 because 2, 5, and 7 are pairwise coprime. For each of the six permutations (p, q, r) of $(2, 5, 7)$, take

$$0 < U_p < U_{pq} < G,$$

where U_d is the unique subgroup of order d. In product notation U_p is the coordinate subgroup for prime p, and U_{pq} is the product of the p and q coordinate subgroups. These are all the series because simple cyclic factors have prime order; the successive factor orders must be a permutation of $2, 5, 7$.

Exercise 35.11: The center of $S_3 \times \mathbb{Z}_4$

An element (g, z) commutes with every (h, w) if and only if $gh = hg$ for every h; the second coordinate already commutes because $\mathbb{Z}_4$ is abelian. Since $Z(S_3) = \{e\}$,

$$Z(S_3 \times \mathbb{Z}_4) = \{e\} \times \mathbb{Z}_4.$$

For example a transposition fails to commute with a different transposition, and a nonidentity 3-cycle fails to commute with a transposition, verifying the stated center of S_3.

Exercise 35.12: The center of $S_3 \times D_4$

Write $D_4 = \langle r, s \rangle$ with $r^4 = s^2 = e$ and $sr = r^{-1}s$. Its center is $\{e, r^2\}$: r and r^3 fail to commute with s, and every reflection fails to commute with r. Commutation in a product is coordinatewise, so

$$Z(S_3 \times D_4) = \{e\} \times \{e, r^2\}.$$

In the textbook's notation $e = \rho_0$ and $r^2 = \rho_2$.

Exercise 35.13: Ascending central series of $S_3 \times \mathbb{Z}_4$

The first nontrivial term is $Z_1 = \{e\} \times \mathbb{Z}_4$. The quotient $G/Z_1 \cong S_3$ has trivial center. Taking the inverse image of that trivial center gives $Z_2 = Z_1$, and the same reasoning repeats. Thus the series, including its initial identity term, is

$$\{(e, 0)\} \leq \{e\} \times \mathbb{Z}_4 \leq \{e\} \times \mathbb{Z}_4 \leq \cdots .$$

It never reaches G. Repeated terms must be retained when describing the ascending central sequence.

Exercise 35.14: Ascending central series of $S_3 \times D_4$

The first term is $Z_1 = \{e\} \times \{e, r^2\}$. Now $G/Z_1 \cong S_3 \times \mathbb{Z}_2^2$, whose center is $\{e\} \times \mathbb{Z}_2^2$. Its inverse image is $Z_2 = \{e\} \times D_4$. Since $G/Z_2 \cong S_3$ has trivial center, every later term equals Z_2. Thus

$$1 \leq \{e\} \times \{e, r^2\} \leq \{e\} \times D_4 \leq \{e\} \times D_4 \leq \cdots .$$

Each term is normal in G, as required for the construction.

Exercise 35.15: Definition of a composition series

The proposed definition is correct, provided "maximal normal" means maximal among *proper* normal subgroups. The finite strict chain starts at $\{e\}$ and ends at G, and each $H_i \trianglelefteq H_{i+1}$ has no normal subgroup strictly between it and H_{i+1}. By the correspondence theorem this is equivalent to H_{i+1}/H_i being simple.

Exercise 35.16: Definition of solvability

Replace "a composition series of abelian groups" by "a composition series whose *factor groups* are abelian." The intermediate groups need not be abelian: $1 < A_3 < S_3$ is a solvable composition series even though S_3 is nonabelian. This uses the textbook's Definition 35.18. It restricts solvability to groups admitting such a finite composition series; the broader modern definition uses a finite subnormal series with abelian factors, allowing infinite groups such as $\mathbb{Z}$.

Exercise 35.17: True or false: group series

(a) True. Normality in G implies normality in every intermediate subgroup.

(b) False. In D_4, $1 < \langle s \rangle < \langle s, r^2 \rangle < D_4$ is subnormal, but $\langle s \rangle$ is not normal in D_4.

(c) True under Definition 35.12 in this text: a principal series is normal with simple factors, hence subnormal with simple factors.

(d) False. The composition series in (b) has order-2 factors but is not normal in D_4.

(e) False. $\mathbb{Z}_6$ has $0 < \langle 2 \rangle < \mathbb{Z}_6$ and $0 < \langle 3 \rangle < \mathbb{Z}_6$; moreover $\mathbb{Z}$ has no composition series.

(f) True. Repeatedly choose a maximal proper normal subgroup; orders strictly decrease, so the process terminates.

(g) False. Simple factors occur in every composition series by definition; S_5 has one but is not solvable, because A_5 is a nonabelian factor.

(h) False. $1 < A_7 < S_7$ has the nonabelian simple factor A_7.

(i) True. The simple factor groups, like primes in an integer factorization, are determined up to order and isomorphism, though the intermediate subgroups need not be unique.

(j) True. A group of prime order is cyclic, so its one factor is abelian and simple.

The textbook's definition of a principal/chief series here is stronger than the usual modern chief-series convention; the answers follow its stated convention.

Exercise 35.18: A composition series of S_3^2

Take

$$1 < A_3 \times 1 < S_3 \times 1 < S_3 \times A_3 < S_3 \times S_3.$$

Each term is normal in the next: this follows from $A_3 \trianglelefteq S_3$ and coordinatewise conjugation. The factors, from bottom to top, are $\mathbb{Z}_3, \mathbb{Z}_2, \mathbb{Z}_3, \mathbb{Z}_2$. They are simple and abelian, so the series is a composition series and $S_3 \times S_3$ is solvable.

Exercise 35.19: Solvability of the square-symmetry group

The chain $1 < \langle r^2 \rangle < \langle r \rangle < D_4$ has subgroup orders $1, 2, 4, 8$. The first subgroup is central, and the rotation subgroup has index 2, hence is normal. Every factor has order 2, so every factor is simple and abelian. Therefore D_4 is solvable. The definition does not require D_4 itself to commute.

Exercise 35.20: Schreier arrays in $\mathbb{Z}_{36}$

Plan. Retain repeated entries while forming the rectangular Schreier arrays. In $\mathbb{Z}_n$, $\langle a \rangle + \langle b \rangle = \langle \gcd(a,b) \rangle$ and $\langle a \rangle \cap \langle b \rangle = \langle \operatorname{lcm}(a,b) \rangle$ when $a, b \mid n$; write $\langle n \rangle = 0$. Use $H_{ij} = H_i + (H_{i+1} \cap K_j)$ and $K_{ji} = K_j + (K_{j+1} \cap H_i)$. The arrays for chains (3) and (4) are respectively

$$\begin{array}{ccccc} 0 & \leq & 0 & \leq & \langle 12 \rangle \\ \langle 12 \rangle & \leq & \langle 6 \rangle & \leq & \langle 3 \rangle \\ \langle 3 \rangle & \leq & \langle 3 \rangle & \leq & G \end{array} \qquad \begin{array}{ccccccc} 0 & \leq & 0 & \leq & \langle 18 \rangle & \leq & \langle 18 \rangle \\ \langle 18 \rangle & \leq & \langle 6 \rangle & \leq & \langle 3 \rangle & \leq & G \end{array}.$$

Read each row left to right, identifying its last term with the first term of the next row. The factor-order array for the first chain is

$$\begin{pmatrix} 1 & 3 \\ 2 & 2 \\ 1 & 3 \end{pmatrix}.$$

The second chain has the transpose array of factor orders: the factor in row i, column j of the first chain corresponds to row j, column i of the second. All factors are cyclic; order 1 denotes a repeated group and thus a trivial factor. Deleting repeated entries gives the requested isomorphic refinements. This also exhibits every factor correspondence, rather than only their unordered lists.

Exercise 35.21: Schreier arrays in $\mathbb{Z}_{24}$

Plan. Retain repeated entries while forming the rectangular Schreier arrays. In $\mathbb{Z}_n$, $\langle a \rangle + \langle b \rangle = \langle \gcd(a,b) \rangle$ and $\langle a \rangle \cap \langle b \rangle = \langle \operatorname{lcm}(a,b) \rangle$ when $a, b \mid n$; write $\langle n \rangle = 0$. Use $H_{ij} = H_i + (H_{i+1} \cap K_j)$ and $K_{ji} = K_j + (K_{j+1} \cap H_i)$. The arrays for chains (3) and (4) are respectively

$$\begin{array}{ccccccc} 0 & \leq & \langle 12 \rangle & \leq & \langle 12 \rangle & \leq & \langle 12 \rangle \\ \langle 12 \rangle & \leq & \langle 12 \rangle & \leq & \langle 12 \rangle & \leq & \langle 4 \rangle \\ \langle 4 \rangle & \leq & \langle 2 \rangle & \leq & G & \leq & G \end{array} \qquad \begin{array}{ccccccc} 0 & \leq & \langle 12 \rangle & \leq & \langle 12 \rangle & \leq & \langle 6 \rangle \\ \langle 6 \rangle & \leq & \langle 6 \rangle & \leq & \langle 6 \rangle & \leq & \langle 3 \rangle \\ \langle 3 \rangle & \leq & \langle 3 \rangle & \leq & G & \leq & G \end{array}.$$

Read each row left to right, identifying its last term with the first term of the next row. The factor-order array for the first chain is

$$\begin{pmatrix} 2 & 1 & 1 \\ 1 & 1 & 3 \\ 2 & 2 & 1 \end{pmatrix}.$$

The second chain has the transpose array of factor orders: the factor in row i, column j of the first chain corresponds to row j, column i of the second. All factors are cyclic; order 1 denotes a repeated group and thus a trivial factor. Deleting repeated entries gives the requested isomorphic refinements. This also exhibits every factor correspondence, rather than only their unordered lists.

Exercise 35.22: Normality in an intersection

Let $x \in H \cap K$ and $y \in H^* \cap K$. Since $x \in H$ and $H^* \trianglelefteq H$, the conjugate $xyx^{-1} \in H^*$. Since $x, y \in K$, it also lies in K. Thus conjugation by x preserves $H^* \cap K$; using x^{-1} yields equality. Therefore $H^* \cap K \trianglelefteq H \cap K$.

Exercise 35.23: Multiplying finite factor orders

Start with $|H_0| = 1$. If $|H_i|$ is finite and $[H_{i+1} : H_i] = s_{i+1}$, then H_{i+1} is the disjoint union of s_{i+1} cosets, each with $|H_i|$ elements. Hence $|H_{i+1}| = s_{i+1}|H_i|$. Induction gives $|H_j| = s_1 \cdots s_j$ for every j. At $j = n$ this proves $|G| = s_1 \cdots s_n$. Normality is needed to call the factors groups, although the coset-counting argument itself only uses subgroup indices.

Exercise 35.24: No infinite abelian composition series

A simple abelian group is cyclic of prime order: a nonidentity element generates a nontrivial normal subgroup, hence the whole group; an infinite cyclic group or a finite cyclic group of composite order has a proper nontrivial subgroup. If an abelian group had a composition series, all its factors would consequently be finite of prime order. Exercise 35.23 would make the group finite. This contradicts the hypothesis that it is infinite.

Exercise 35.25: Finite products of solvable groups

For two factors choose composition series $1 = H_0 < \cdots < H_r = H$ and $1 = K_0 < \cdots < K_s = K$ with abelian factors. Form

$$1 < H_1 \times 1 < \cdots < H \times 1 < H \times K_1 < \cdots < H \times K.$$

Normality holds coordinatewise. The factors in the first portion are H_i/H_{i-1}; those in the second are K_j/K_{j-1}. Thus every factor is simple and abelian, so this is a solvable composition series. Apply the two-factor construction repeatedly for any finite number of groups.

Exercise 35.26: Subgroups of solvable groups

Let $1 = H_0 < \cdots < H_n = G$ have abelian simple factors, hence cyclic factors of prime order. Set $K_i = K \cap H_i$. For $x \in K_i$, conjugation preserves $K \cap H_{i-1}$, so $K_{i-1} \trianglelefteq K_i$. The map $K_i \to H_i/H_{i-1}$, $x \mapsto xH_{i-1}$, has kernel K_{i-1}. Therefore K_i/K_{i-1} is isomorphic to a subgroup of a prime-order cyclic group. It is either trivial or that entire prime-order group. After deleting repetitions the intersections form a composition series of K with abelian factors. Thus K is solvable. The prime-order property is essential: intersections with an arbitrary composition series need not have simple factors.

Exercise 35.27: Inserting a simple normal subgroup

Because $N \trianglelefteq G$, each H_iN is a subgroup. Also $H_{i-1}N \trianglelefteq H_iN$: conjugation by H_i preserves both H_{i-1} and N, while conjugation by $n \in N$ fixes every coset modulo N and thus preserves $H_{i-1}N$. The composite map

$$H_i \longrightarrow H_iN/H_{i-1}N$$

is onto, with kernel $H_i \cap (H_{i-1}N)$ containing H_{i-1}. Hence its target is a quotient of the simple group H_i/H_{i-1}, and is either trivial or simple. Prepend 1 to $N = H_0N \leq H_1N \leq \cdots \leq G$. Its initial factor is the assumed simple group N; all later nontrivial factors are simple by the preceding argument. Delete repeated terms to obtain the required composition series.

Exercise 35.28: Composition series descend to quotients

Let $\gamma : G \to G/N$ be the quotient map. For $x \in H_i$ and $y \in H_{i-1}$, $\gamma(x)\gamma(y)\gamma(x)^{-1} = \gamma(xyx^{-1}) \in \gamma(H_{i-1})$. Thus the image terms form a subnormal chain. The onto map

$$H_i/H_{i-1} \longrightarrow \gamma(H_i)/\gamma(H_{i-1}), \qquad hH_{i-1} \longmapsto \gamma(h)\gamma(H_{i-1})$$

is well defined, since H_{i-1} maps into $\gamma(H_{i-1})$. A quotient of a simple group is either trivial or simple. The image chain starts at 1 and ends at G/N; deleting its repetitions therefore produces a composition series of G/N.

Exercise 35.29: Homomorphic images preserve solvability

Let $f : G \to Q$ be a homomorphism, and use a composition series of G with cyclic prime-order factors. Repeat the argument of Exercise 35.28 with the onto map $G \to f(G)$: each successive image factor is a quotient of a cyclic prime-order group, hence either trivial or cyclic of that prime order. Delete repetitions. The result is a composition series of $f(G)$ with abelian factors, proving that the image is solvable. No surjectivity onto the originally named codomain Q is needed; the assertion concerns $f(G)$.

36 Sylow Theorems

For a finite group of order $p^a m$, where $p \nmid m$, a Sylow p-subgroup has order p^a; its number n_p divides m and satisfies $n_p \equiv 1 \pmod{p}$. A unique Sylow subgroup is normal because conjugation preserves its order.

Exercise 36.1: The Sylow 3-order when $|G| = 12$

Factor $12 = 2^2 \cdot 3$. A Sylow 3-subgroup has the largest power of 3 dividing $|G|$, namely $3^1 = 3$, as its order. This is the subgroup's order, not the number of such subgroups.

Exercise 36.2: The Sylow 3-order when $|G| = 54$

Since $54 = 2 \cdot 3^3$, the largest power of 3 dividing 54 is 27. Hence every Sylow 3-subgroup has order 27.

Exercise 36.3: Counting Sylow 2-subgroups for order 24

Sylow's counting theorem gives $n_2 \mid 24/8 = 3$ and $n_2 \equiv 1 \pmod{2}$. Both divisors 1 and 3 satisfy these conditions, so the blanks are 1 and 3. The theorem gives possible values, without asserting that both occur in a particular group.

Exercise 36.4: Counting Sylow subgroups for order 255

For $p = 3$, $n_3 \mid 85$ and $n_3 \equiv 1 \pmod{3}$. Of $1, 5, 17, 85$, only $1, 85$ qualify. For $p = 5$, $n_5 \mid 51$ and $n_5 \equiv 1 \pmod{5}$; of $1, 3, 17, 51$, only $1, 51$ qualify. Thus the requested alternatives are $n_3 = 1$ or 85, and $n_5 = 1$ or 51. Only the counting theorem is being used here, as requested.

Exercise 36.5: Sylow 3-subgroups of S_4

A Sylow 3-subgroup has order 3, since $24 = 3 \cdot 8$. Every nonidentity element in one is a 3-cycle; each 3-cycle and its inverse generate the same subgroup. The full list is

$$\langle (123) \rangle, \quad \langle (124) \rangle, \quad \langle (134) \rangle, \quad \langle (234) \rangle.$$

There are eight 3-cycles, two per subgroup, so nothing is missing. Conjugation relabels a cycle: $\sigma(ijk)\sigma^{-1} = (\sigma(i)\,\sigma(j)\,\sigma(k))$. Starting from $\langle (123) \rangle$, conjugate by (34), (234), and (1234) to get the other three subgroups respectively. Thus all four are conjugate explicitly.

Exercise 36.6: Two conjugate Sylow 2-subgroups of S_4

Take $P = \langle (1234), (13) \rangle$. With $r = (1234)$, $s = (13)$, we have $r^4 = s^2 = e$ and $srs = r^{-1}$. Its eight distinct elements are

$$e, (1234), (13)(24), (1432), (13), (24), (14)(23), (12)(34).$$

Since 8 is the largest power of 2 dividing 24, P is Sylow. Let $\tau = (34)$ and set $Q = \tau P \tau^{-1} = \langle (1243), (14) \rangle$. It again has eight elements and is Sylow. It differs from P because $(1243) \notin P$. The displayed equation itself demonstrates that P and Q are conjugate, rather than merely appealing to the general theorem.

Exercise 36.7: Definition of a p-group

An element need not have order exactly p. The correct definition is: G is a p-group if every element has order a power p^k with $k \geq 0$ (the identity has order $p^0 = 1$). For finite G this is equivalent to $|G|$ being a power of p. For example $\mathbb{Z}_{p^2}$ is a p-group and has an element of order p^2.

Exercise 36.8: Definition of the normalizer

The normalizer is a set of *elements of* G, not a set of automorphisms:

$$N_G(H) = \{g \in G : gHg^{-1} = H\}.$$

Equivalently it consists of those elements whose induced inner automorphism preserves H. Distinct group elements can induce the same inner automorphism, so replacing elements by automorphisms changes the set and potentially its order.

Exercise 36.9: Definition of a Sylow subgroup

For a finite group with $|G| = p^a m$ and $p \nmid m$, a Sylow p-subgroup is a subgroup of order p^a. Equivalently it is a maximal p-subgroup under inclusion. Say *a* Sylow subgroup, since there may be several. "The largest" suggests a unique subgroup containing all others, which usually does not exist.

Exercise 36.10: True or false: Sylow theory

(a) True: this is Sylow conjugacy.

(b) True: $n_5 \mid 3$ and $n_5 \equiv 1 \pmod 5$ force $n_5 = 1$.

(c) True by the definition of a Sylow p-subgroup.

(d) False: $\{0, 2\} < \mathbb{Z}_4$ is a 2-subgroup but is not Sylow; the entire group has order 4.

(e) True: conjugation is trivial in an abelian group, and Sylow subgroups are conjugate.

(f) False: in S_3, the normalizer of $\langle (12) \rangle$ equals $\langle (12) \rangle$, which is not normal in S_3.

(g) True: every element of the normalizer conjugates H onto itself.

(h) True: uniqueness implies conjugation invariance, and normality combined with Sylow conjugacy implies uniqueness.

(i) False: in an abelian group $N_G(H) = G$, which can strictly contain H.

(j) False: the whole group is its Sylow p-subgroup.

Exercise 36.11: The normalizer is a subgroup

Certainly $eHe^{-1} = H$. If $aHa^{-1} = H$ and $bHb^{-1} = H$, then $(ab)H(ab)^{-1} = a(bHb^{-1})a^{-1} = H$. Also $aHa^{-1} = H$ implies $a^{-1}Ha = H$, by conjugating the equality with a^{-1}. Hence the normalizer contains the identity and is closed under products and inverses, so it is a subgroup.

Exercise 36.12: A unique proper Sylow subgroup is normal

Let P be the only Sylow p-subgroup. For each $g \in G$, conjugation is an isomorphism, so gPg^{-1} has the same order as P and is again Sylow. Uniqueness gives $gPg^{-1} = P$. Thus $P \trianglelefteq G$. It is nontrivial because $p \mid |G|$, and proper because a distinct prime q divides $|G|$. This is a nontrivial proper normal subgroup, so G is not simple.

Exercise 36.13: A normal order-9 subgroup for order 45

Since $45 = 3^2 \cdot 5$, Sylow 3-subgroups have order 9. Their number divides 5 and is 1 modulo 3. Of the divisors 1 and 5, only 1 is 1 modulo 3. Thus there is a unique Sylow 3-subgroup; conjugation preserves its order and hence preserves it. This is the required normal subgroup of order 9.

Exercise 36.14: Finite p-groups and prime-power orders

If $|G| = p^a$, Lagrange's theorem makes every element order divide p^a, so each order is a power of p. Conversely suppose every element order is a power of p. If a different prime q divided $|G|$, Cauchy's theorem would produce an element of order q, contradicting the assumption. Thus no prime other than p divides $|G|$, so $|G| = p^a$. The trivial group corresponds to $a = 0$.

Exercise 36.15: A Sylow normalizer is self-normalizing

Put $M = N_G(P)$. Then $P \trianglelefteq M$, and P remains a Sylow subgroup of M because $P \subseteq M \subseteq G$ leaves no room for a larger power of p. Hence P is the unique Sylow p-subgroup of M. If $g \in N_G(M)$, conjugation by g is an automorphism of M and carries P to another Sylow p-subgroup of M. Uniqueness forces $gPg^{-1} = P$, so $g \in M$. This proves $N_G(M) \subseteq M$. Every subgroup normalizes itself, giving the reverse inclusion. Therefore $N_G(N_G(P)) = N_G(P)$.

Exercise 36.16: Conjugating a p-subgroup into a fixed Sylow subgroup

Let the p-group H act by left multiplication on the left-coset set G/P. Its size $[G : P]$ is not divisible by p. The orbit congruence therefore implies that the number of cosets fixed by all of H is nonzero modulo p, hence positive. Choose a fixed coset xP. For every $h \in H$, $hxP = xP$, so $x^{-1}hx \in P$. Thus $x^{-1}Hx \subseteq P$. Taking $g = x^{-1}$ yields the requested $gHg^{-1} \subseteq P$.

Exercise 36.17: A normal order-125 subgroup

The group order is $(35)^3 = 5^3 \cdot 7^3$. A Sylow 5-subgroup has order 125. Its number divides 343, whose divisors are $1, 7, 49, 343$; their residues modulo 5 are $1, 2, 4, 3$. Only 1 is allowed by Sylow's congruence.

Therefore the order-125 subgroup is unique and hence normal.

Exercise 36.18: Nonsimplicity at order 255

A Sylow 17-subgroup has order 17. Its number divides 15 and is 1 modulo 17. The divisors $1, 3, 5, 15$ include only one such number, namely 1. Thus the subgroup is unique, normal, nontrivial, and proper. Consequently no group of order 255 is simple.

Exercise 36.19: The small-cofactor criterion and its prime exception

Qualification of the printed statement. As written it includes $m = 1, r = 1$, so it is false: $\mathbb{Z}_p$ is simple and has order $p^1 \cdot 1$ with $1 < p$. For the intended composite-order assertion, let $|G| = p^r m$ with $1 < m < p$. Since $p \nmid m$, a Sylow p-subgroup has order p^r. Its number divides m and is congruent to 1 modulo p. But every positive divisor of m lies between 1 and $m < p$, so only 1 is possible. The unique Sylow subgroup is nontrivial, normal, and proper. If $m = 1$ and $r > 1$, the group is a p-group. Its center is nontrivial by Exercise 36.20. If the center is proper it is the required normal subgroup; if the group is abelian, any order-p subgroup is normal and proper. Thus only the prime-order case must be excluded.

Exercise 36.20: Conjugation fixed points and the center

(a) A point $x \in G$ is fixed by every conjugation precisely when $gxg^{-1} = x$ for every g, equivalently $gx = xg$ for every g. Thus the fixed-point set is $Z(G)$. **(b)** When $|G| = p^n$ with $n \geq 1$, the orbit congruence for this action gives $|Z(G)| \equiv |G| \equiv 0 \pmod{p}$. The center contains e, so its order is a positive multiple of p and in particular at least p. Hence it is nontrivial.

Exercise 36.21: Normal subgroups of every prime-power order

We prove the stronger assertion of a nested normal chain by induction on n. For $n = 0$, take $H_0 = G = 1$. For $n \geq 1$, the center is nontrivial, so Cauchy's theorem gives a central subgroup C of order p. Centrality makes $C \trianglelefteq G$. The quotient G/C has order p^{n-1}. By induction it has a nested chain of subgroups $\overline{H}_j \trianglelefteq G/C$ of orders p^j for $0 \leq j \leq n-1$. If $\pi : G \to G/C$, set $H_0 = 1$ and $H_{j+1} = \pi^{-1}(\overline{H}_j)$. Preimages of normal subgroups are normal, and each preimage has order $|C|\,|\overline{H}_j| = p^{j+1}$. They are nested because the quotient terms are nested. This gives all required H_i, with $H_n = G$.

Exercise 36.22: Normal p-subgroups lie in every Sylow subgroup

Let S be any Sylow p-subgroup. Since $P \trianglelefteq G$, PS is a subgroup. Its order is

$$|PS| = \frac{|P||S|}{|P \cap S|},$$

a power of p, since all three orders are powers of p. It contains S, which already has the largest possible p-power order in G. Thus $PS = S$, and hence $P \subseteq S$. Since S was arbitrary, this holds for every Sylow p-subgroup.

37 Applications of the Sylow Theory

Exercise 37.1: Conjugacy classes of the square-symmetry group

(a) Write $D_4 = \{e, r, r^2, r^3, s, rs, r^2s, r^3s\}$ with $srs = r^{-1}$. Conjugating a rotation by s reverses it; conjugating a reflection by r advances its exponent by 2. The classes are therefore

$$\{e\}, \quad \{r^2\}, \quad \{r, r^3\}, \quad \{s, r^2s\}, \quad \{rs, r^3s\}.$$

These are disjoint and account for all eight elements. In the textbook's labels the last two classes are the two axis reflections and the two diagonal reflections, while $r = \rho_1$. **(b)** Their sizes give the class equation $8 = 1 + 1 + 2 + 2 + 2$. The two singleton classes are exactly the center $\{e, r^2\}$.

Exercise 37.2: Composite orders below 60

Plan. For completeness, organize the finite list instead of making an unsupported assertion for each order. If a Sylow number is 1, its subgroup is normal. A nontrivial p-group has nontrivial center and, if its order is composite, is not simple. The prime-power composite orders are

$$4, 8, 9, 16, 25, 27, 32, 49.$$

For the following remaining orders, the indicated prime has a unique Sylow subgroup, directly from $n_p \mid |G|/p^a$ and $n_p \equiv 1 \pmod p$:

p	$\lvert G\rvert$
3	$6, 18, 54$
5	$10, 15, 20, 35, 40, 45, 50$
7	$14, 21, 28, 42$
11	$22, 33, 44$
13	$26, 39, 52$
17	$34, 51$
19	$38, 57$
23	46
29	58
11	55

(The rows may use any prime giving uniqueness.) This leaves $12, 24, 30, 36, 48, 56$.

- For 24 and 48, n_2 is 1 or 3. If it is 3 and G were simple, its nontrivial conjugation action on the three Sylow subgroups would embed G into S_3, impossible by orders.
- For 12 and 36, n_3 is 1 or 4. In the latter case simplicity would embed G into S_4. The sign map must then be trivial (a simple group of order greater than 2 has no nontrivial map to $\mathbb{Z}_2$), so G embeds into A_4. Order 36 is too large; at order 12 this would give $G \cong A_4$, which has its normal order-4 subgroup of double transpositions.
- For 30, unless a Sylow 3- or 5-subgroup is normal, there are ten order-3 subgroups and six order-5 subgroups. Their nonidentity elements are disjoint, requiring $10 \cdot 2 + 6 \cdot 4 = 44 > 29$ nonidentity elements.

- For 56, n_7 is 1 or 8. In the latter case the order-7 subgroups already occupy 48 nonidentity elements. Only seven nonidentity elements remain. Every Sylow 2-subgroup has exactly seven nonidentity elements, all among these remaining ones; hence all such subgroups coincide and the Sylow 2-subgroup is normal.

This exhausts every composite order below 60. The conjugation actions used above are nontrivial and, under the assumption of simplicity, have trivial kernel, justifying the embeddings.

Exercise 37.3: True or false: applications of Sylow theory

(a) True. $159 = 3 \cdot 53$, and $53 \not\equiv 1 \pmod 3$; the two-prime criterion gives a cyclic group. Explicitly $n_{53} = 1$ and $n_3 \mid 53$, $n_3 \equiv 1 \pmod 3$ force $n_3 = 1$, so the normal factors commute and their product is cyclic.

(b) True. $102 = 2 \cdot 3 \cdot 17$, and $n_{17} \mid 6$, $n_{17} \equiv 1 \pmod{17}$ force a normal order-17 subgroup.

(c) False. $\mathbb{Z}_6$ is solvable but 6 is not a prime power.

(d) True. The normal chain of Exercise 36.21 has prime-order factors, so a finite p-group is solvable.

(e) This is a judgment about the method rather than a formal theorem; "true" describes the intended observation. Individual Sylow counts and class equations for many distinct orders become lengthy, and some cases require additional arguments.

(f) True. For order 21, $n_7 \mid 3$ and $n_7 \equiv 1 \pmod 7$ give $n_7 = 1$.

(g) True. Its center has order a positive multiple of 5 by the class equation for 5-groups, so it contains at least five elements.

(h) True. For order 42, $n_7 \mid 6$ and $n_7 \equiv 1 \pmod 7$ give $n_7 = 1$.

(i) False. A subgroup order must divide 42, but $8 \nmid 42$.

(j) False. There are other simple groups, for example $\mathrm{GL}(3, \mathbb{F}_2)$ of order $(8-1)(8-2)(8-4) = 168$, which is not of prime order or the order of an alternating group. One way to check simplicity is to count its conjugacy classes: their sizes are $1, 21, 56, 42, 24, 24$ (the nonidentity element orders are respectively $2, 3, 4, 7, 7$). No union of these classes containing the identity has a proper nontrivial size dividing 168. A normal subgroup is a union of conjugacy classes and its order divides 168, so none exists. The class sizes follow from centralizer orders $8, 3, 4, 7, 7$, respectively, by counting invertible matrices commuting with representatives.

Exercise 37.4: Groups of order $5 \cdot 7 \cdot 47$

Apply Sylow to each prime. The possible counts satisfy

$$n_{47} \mid 35, \quad n_{47} \equiv 1 \pmod{47}; \qquad n_7 \mid 235, \quad n_7 \equiv 1 \pmod 7.$$

The divisors of 35 are all below 47, so $n_{47} = 1$. The divisors of $235 = 5 \cdot 47$ are $1, 5, 47, 235$, with residues $1, 5, 5, 4$ modulo 7, so $n_7 = 1$. Also $n_5 \mid 329$; its possible divisors $1, 7, 47, 329$ have residues $1, 2, 2, 4$ modulo 5, forcing $n_5 = 1$. Thus all three Sylow subgroups are normal, cyclic, and of pairwise coprime orders. Two distinct ones commute: a commutator of their elements belongs to both normal

subgroups, whose intersection is trivial by Lagrange's theorem. Their product has all $5 \cdot 7 \cdot 47$ elements and is the direct product of three cyclic groups of pairwise coprime orders. Hence it is cyclic and therefore abelian.

Exercise 37.5: No simple group of order 96

A Sylow 2-subgroup has order 32, and $n_2 \mid 3$, so $n_2 = 1$ or 3. If $n_2 = 1$, it is a proper nontrivial normal subgroup. If $n_2 = 3$, conjugation on the three Sylow subgroups defines a nontrivial homomorphism $G \to S_3$. Were G simple, the kernel, a normal subgroup, would be trivial because the action is nontrivial. This would embed 96 elements into the six-element group S_3, impossible. Thus the kernel is itself a nontrivial proper normal subgroup, proving nonsimplicity in this case too.

Exercise 37.6: No simple group of order 160

Since $160 = 2^5 \cdot 5$, the Sylow 2-number divides 5 and is odd, so it is 1 or 5. The first case gives a normal subgroup of order 32. In the second case the conjugation action gives a nontrivial homomorphism $G \to S_5$. Its kernel cannot be trivial, because an injective map would require $160 \le |S_5| = 120$; and it is not all of G, since the action is nontrivial. Thus it is a nontrivial proper normal subgroup. Both cases exclude simplicity.

Exercise 37.7: An order-15 subgroup in every order-30 group

The counting argument of Example 37.12 shows that an order-30 group has a normal subgroup N of order 3 or of order 5: otherwise ten order-3 subgroups and six order-5 subgroups would require 44 nonidentity elements. If $|N| = 3$, the quotient G/N has order 10, and Cauchy's theorem gives a subgroup Q of order 5. Its preimage under the quotient map has order $3 \cdot 5 = 15$. If $|N| = 5$, the quotient has order 6 and has an order-3 subgroup; its preimage has order $5 \cdot 3 = 15$. Preimages of subgroups are subgroups, so either alternative proves the assertion. The order computation counts $|N|$ elements in each coset of the preimage.

Exercise 37.8: Cycle type and partitions

(a) The permutation $\tau\sigma\tau^{-1}$ sends $\tau(a_i)$ to $\tau(a_{i+1})$, with the last sent to the first, and fixes points outside this cycle. Thus it is $(\tau(a_1) \ldots \tau(a_m))$. **(b)** For two cycles of the same length, define τ to send their entries in cyclic order to one another and extend this bijection arbitrarily to the other points. Part (a) gives the desired conjugation. **(c)** For products of disjoint cycles with the same lengths, do the same on each disjoint support and combine these bijections. Conjugation respects products, so it sends the first product to the second. **(d)** Conversely conjugation only relabels entries and therefore preserves all cycle lengths, including fixed points as length-1 cycles. Thus conjugacy classes correspond exactly to unordered lists of positive lengths summing to n, namely partitions of n. **(e)** The numbers for $n = 1, 2, 3, 4, 5, 6, 7$ are

$$1, 2, 3, 5, 7, 11, 15.$$

For example a systematic way to compute them is to let $P(n, k)$ count partitions with parts at most k and use $P(n, k) = P(n, k-1) + P(n-k, k)$, with $P(0, k) = 1$ and negative arguments zero. The two terms distinguish partitions having no part k from those obtained by removing one part k.

Exercise 37.9: Conjugacy classes and class equation of S_4

By Exercise 37.8 there is one class for each cycle type. A class with m_k cycles of length k has size

$$\frac{n!}{\prod_k k^{m_k} m_k!}.$$

To see the denominator, rotate the entries within each length-k cycle in k ways and permute equal-length cycles in $m_k!$ ways; these do not change the permutation. For S_4 the classes are

type	representative	size
1^4	e	1
$2\,1^2$	(12)	6
2^2	$(12)(34)$	3
$3\,1$	(123)	8
4	(1234)	6

Here each row means all permutations of that type. Hence the class equation is $24 = 1 + 6 + 3 + 8 + 6$.

Exercise 37.10: Class equations of S_5 and S_6

Use the cycle-type formula $n!/\prod_k k^{m_k} m_k!$ from Exercise 37.9. For S_5 the complete table is

type	1^5	$2\,1^3$	$2^2\,1$	$3\,1^2$	$3\,2$	$4\,1$	5
size	1	10	15	20	20	30	24

Thus $120 = 1 + 10 + 15 + 20 + 20 + 30 + 24$. For S_6, list the eleven partitions as follows:

type	size	type	size
1^6	1	$2\,1^4$	15
$2^2\,1^2$	45	2^3	15
$3\,1^3$	40	$3\,2\,1$	120
3^2	40	$4\,1^2$	90
$4\,2$	90	$5\,1$	144
6	120		

Consequently

$$720 = 1 + 15 + 45 + 15 + 40 + 120 + 40 + 90 + 90 + 144 + 120.$$

The displayed sums equal the group orders, checking that all types and counts have been included.

Exercise 37.11: Conjugacy classes and abelian p-groups

The structure theorem writes an abelian group of order p^n uniquely up to isomorphism as

$$\mathbb{Z}_{p^{r_1}} \times \cdots \times \mathbb{Z}_{p^{r_t}}, \quad r_i \geq 1, \quad r_1 \leq \cdots \leq r_t, \quad \sum r_i = n.$$

The unordered exponent list is a partition of n, and different lists give nonisomorphic groups by uniqueness of the prime-power decomposition. Thus there are $p(n)$ isomorphism types of abelian groups of order p^n. Exercise 37.8 gives exactly $p(n)$ conjugacy classes in S_n, since cycle lengths provide the same partitions. The shared count comes from the common partition parameter, not from an asserted group isomorphism.

Exercise 37.12: The center of S_n for $n > 2$

Let $\sigma \neq e$, choose i with $\sigma(i) = j \neq i$, and choose k different from both i, j; this is possible because $n > 2$. Conjugation gives

$$\sigma(i\ k)\sigma^{-1} = (j\ \sigma(k)).$$

This transposition cannot equal $(i\ k)$, because its support contains j, which is neither i nor k. Therefore σ does not commute with $(i\ k)$ and is not central. Every nonidentity permutation fails centrality, while the identity commutes with everything. Hence $Z(S_n) = \{e\}$.

38 Free Abelian Groups

Exercise 38.1: A basis of $\mathbb{Z}^3$ with no zero coordinates

Take $u = (1, 1, 1)$, $v = (1, 2, 2)$, $w = (1, 2, 3)$. For any $(e, f, g) \in \mathbb{Z}^3$, solving $xu + yv + zw = (e, f, g)$ by subtracting coordinates gives

$$z = g - f, \qquad y = 2f - e - g, \qquad x = 2e - f.$$

These are integers and substitution recovers all three coordinates. The formulas also give a unique coefficient triple, so the vectors generate $\mathbb{Z}^3$ and are independent over $\mathbb{Z}$. Thus they form a basis, with every coordinate nonzero.

Exercise 38.2: The vectors $(2, 1)$ and $(3, 1)$

Yes. For an arbitrary (e, f), the equations $2x + 3y = e$, $x + y = f$ give

$$y = e - 2f, \qquad x = 3f - e.$$

Both coefficients are integers, so every integer pair is generated. They are uniquely determined, so the only representation of zero is $x = y = 0$. These are precisely the generating and independence requirements for a basis of a free abelian group.

Exercise 38.3: The vectors $(2, 1)$ and $(4, 1)$

No. Any integer combination $x(2, 1) + y(4, 1) = (2x + 4y, x + y)$ has even first coordinate. Therefore $(1, 0)$ is not generated. The two vectors are linearly independent over $\mathbb{R}$ (their determinant is -2), but that does not make them a $\mathbb{Z}$-basis: integer coefficients, rather than arbitrary real coefficients, are required.

Exercise 38.4: The determinant criterion for an integer basis

Put $D = ad - bc$, the determinant of the matrix with columns (a, b) and (c, d). If $D = \pm 1$, solving $x(a, b) + y(c, d) = (e, f)$ gives

$$x = \frac{de - cf}{D}, \qquad y = \frac{af - be}{D}.$$

These coefficients are integers and unique for all $e, f \in \mathbb{Z}$, proving the basis property. Conversely, if the two vectors are a basis, there is an integer matrix B whose columns express $(1, 0), (0, 1)$ in that basis. For $A = \begin{pmatrix} a & c \\ b & d \end{pmatrix}$ this gives $AB = I$. Taking determinants yields $D \det B = 1$. Both factors are integers, so $D = \pm 1$. Hence the necessary and sufficient condition is $ad - bc = \pm 1$.

Exercise 38.5: Definition of rank

Replace “a generating set” by “a basis.” The rank is the cardinality of any basis of the free abelian group. Generating sets can contain redundant elements: $\{1\}$ and $\{1, 2\}$ both generate $\mathbb{Z}$ but have different sizes. Bases have invariant cardinality.

Exercise 38.6: Definition of an abelian basis

The proposed definition is correct. The zero relation with distinct basis elements forces every integer coefficient to vanish, and the set generates the group. The reverse implication in the independence condition is automatic, since a sum with all coefficients zero is zero. A group having such a basis is free abelian.

Exercise 38.7: A proper subgroup of the same rank

For any positive integer r, take $G = \mathbb{Z}^r$ and $K = 2\mathbb{Z} \times \mathbb{Z}^{r-1}$. It is proper since $(1, 0, \ldots, 0) \notin K$. The r vectors $2e_1, e_2, \ldots, e_r$ generate K and are independent over $\mathbb{Z}$, so K has rank r, just as G does. In particular already $2\mathbb{Z} < \mathbb{Z}$ gives a rank-1 example.

Exercise 38.8: True or false: free abelian groups

(a) True. If a multiple of an element is zero, uniqueness of basis coefficients makes every coefficient zero.

(b) True by the structure theorem: the torsion cyclic factors vanish, leaving $\mathbb{Z}^r$.

(c) True: $\mathbb{Z}^r$ has the standard r-element basis.

(d) True. If G has Betti number r and is generated by r elements, write $G \cong \mathbb{Z}^r \oplus T$. A prime $p \mid |T|$ would give $G/pG \cong \mathbb{F}_p^r \oplus T/pT$ of dimension greater than r, impossible with r generators. Thus $T = 0$.

(e) True. The larger set still contains every generator in X, so still generates all of G.

(f) False. $\{1\}$ is a basis of $\mathbb{Z}$, but $\{1, 2\}$ is dependent: $2 \cdot 1 - 1 \cdot 2 = 0$.

(g) False. $\mathbb{Z}$ has only two bases, $\{1\}$ and $\{-1\}$.

(h) True. If $x_1, x_2, \ldots$ is a basis, replacing x_1 by $x_1 + kx_2$ gives a different basis for each $k \in \mathbb{Z}$, since the old x_1 is recovered by subtracting kx_2.

(i) True: a subgroup of $\mathbb{Z}^r$ is free abelian, as proved in this section.

(j) False. For $G = \mathbb{Z}$, $K = 2\mathbb{Z}$, the quotient is $\mathbb{Z}_2$, which has nonzero torsion and is not free abelian.

Exercise 38.9: The coordinate isomorphism

For a basis $x_1, \ldots, x_r$, define $\phi(\sum n_i x_i) = (n_1, \ldots, n_r)$. Uniqueness of a basis expansion makes this well defined, including the zero vector. If $a = \sum n_i x_i$ and $b = \sum m_i x_i$, then $a + b = \sum (n_i + m_i) x_i$, so $\phi(a + b) = \phi(a) + \phi(b)$. If $\phi(a) = 0$, all coefficients are zero, hence $a = 0$; thus the map is injective. Given any integer tuple $(n_1, \ldots, n_r)$, its preimage is $\sum n_i x_i$, so the map is onto. This completes the claimed isomorphism $G \cong \mathbb{Z}^r$.

Exercise 38.10: A free abelian group has no nonzero torsion

Let $a = \sum_{i=1}^{r} n_i x_i$ in a basis, with finitely many coefficients involved, and suppose $ma = 0$ for an integer $m \neq 0$. Then $\sum_i (mn_i)x_i = 0$. Independence forces $mn_i = 0$ in $\mathbb{Z}$ for each i. Since $m \neq 0$, every $n_i = 0$, so $a = 0$. Thus a nonzero element cannot have finite order; the argument works even for an infinite basis because each expansion is finite.

Exercise 38.11: Products of free abelian groups

If X is a basis of G and Y a basis of G', use

$$\{(x, 0) : x \in X\} \ \cup \ \{(0, y) : y \in Y\}.$$

It generates $G \times G'$ by expanding each coordinate separately. A zero integer combination has zero first coordinate, forcing all X-coefficients to be zero, and zero second coordinate, forcing all Y-coefficients to be zero. Thus the displayed set is a basis, and the product is free abelian. This is a finite product of groups; even when the bases are infinite, each individual expansion remains finite.

Exercise 38.12: Finitely generated torsion-free groups

A free abelian group of finite rank has a finite basis and hence is finitely generated; Exercise 38.10 proves it is torsion free. Conversely the structure theorem writes a finitely generated abelian group as $\mathbb{Z}^r \oplus T$, where T is finite. If there is no nonzero element of finite order, T must be trivial. Thus the group is $\mathbb{Z}^r$, which has its standard basis. The zero group is included by the usual empty-basis/rank-zero convention; if "free abelian" is restricted to nonzero groups as in the section's initial definition, state the equivalence for nonzero groups.

Exercise 38.13: Why $\mathbb{Q}$ is not free abelian

Two nonzero rational numbers $x = a/b$ and $y = c/d$ satisfy $(bc)x - (ad)y = 0$, with both coefficients nonzero when $a, c \neq 0$. Thus no independent set over $\mathbb{Z}$ can contain two distinct nonzero rational numbers; zero cannot be a basis element either. A basis for the nonzero group $\mathbb{Q}$ would therefore contain just one nonzero rational q. But its generated subgroup consists of integer multiples of q, and $q/2$ is not one of them. Hence one element cannot generate $\mathbb{Q}$, so $\mathbb{Q}$ is not free abelian.

Exercise 38.14: The p-primary torsion subgroup

The set T_p contains zero. If $p^a x = 0$ and $p^b y = 0$, let $c = \max(a, b)$; then $p^c(x - y) = 0$. Hence $x - y$ has order a power of p (or is zero), and lies in T_p. The one-step subgroup criterion proves $T_p \leq T$. The commutativity of T is used when distributing multiplication by p^c over a difference.

Exercise 38.15: Locating the p-primary factors

In a prime-power product decomposition of T, an element is killed by a power of p exactly when all its coordinates in q-power factors with $q \neq p$ vanish. Indeed, multiplication by p is invertible on a cyclic q-group, because $\gcd(p, q) = 1$. Its repeated powers therefore cannot kill a nonzero coordinate.

Conversely the remaining p-power coordinates are all killed by some common power of p. Restricting the product isomorphism to T_p thus identifies it exactly with the product of the cyclic p-power factors.

Exercise 38.16: Elements killed by an integer

The multiplication map $\mu_n : G \to G$, $x \mapsto nx$, is a homomorphism because G is abelian. Therefore $G[n] = \ker \mu_n$ is a subgroup. Directly, $n0 = 0$, and if $nx = ny = 0$, then $n(x - y) = nx - ny = 0$, proving the same fact without invoking kernels. For a general nonabelian group the power map need not be a homomorphism; the abelian assumption is essential.

Exercise 38.17: The p-torsion of a cyclic p-group

In $\mathbb{Z}_{p^r}$ the equation $px = 0$ means $p^r \mid px$, equivalently $p^{r-1} \mid x$. Thus

$$\mathbb{Z}_{p^r}[p] = \{0, p^{r-1}, 2p^{r-1}, \ldots, (p-1)p^{r-1}\}.$$

It is cyclic generated by p^{r-1} and has order p, so the map $\mathbb{Z}_p \to \mathbb{Z}_{p^r}[p]$ sending k to kp^{r-1} is a well-defined bijective homomorphism.

Exercise 38.18: Counting p-torsion in a product

An element of a direct product is killed by p exactly when every coordinate is killed by p. Hence

$$\left(\prod_{i=1}^{m} \mathbb{Z}_{p^{r_i}}\right)[p] = \prod_{i=1}^{m} (\mathbb{Z}_{p^{r_i}}[p]) \cong \mathbb{Z}_p^m$$

by Exercise 38.17. In particular its size is p^m; this size is intrinsic to the group, independent of a chosen decomposition.

Exercise 38.19: Uniqueness of the p-power exponents

(a) Apply the intrinsic construction $T_p[p]$ to the two decompositions. Their orders are p^m and p^n, so $m = n$. **(b)** Arrange exponents increasingly. Suppose that $r_i = s_i$ for $i < j$ and, seeking a contradiction, $r_j < s_j$. The group $p^{r_j}T_p$ has from the first decomposition one nonzero cyclic factor for each $r_i > r_j$, at most $m - j$ factors. From the second decomposition every index $i \geq j$ has $s_i \geq s_j > r_j$, giving at least $m - j + 1$ nonzero factors; the indices before j have the same behavior in both decompositions. Part (a), applied to this intrinsic subgroup, forbids different numbers of nonzero p-power cyclic factors. Thus $r_j < s_j$ is impossible. Interchange the decompositions to rule out $s_j < r_j$. Start at $j = 1$ and induct to obtain $r_i = s_i$ for all i.

Exercise 38.20: From invariant factors to prime powers

Factor each torsion coefficient m_i into pairwise coprime prime powers. The Chinese remainder isomorphism gives

$$\mathbb{Z}_{m_i} \cong \prod_{p \mid m_i} \mathbb{Z}_{p^{v_p(m_i)}}.$$

Taking the product over i and regrouping by prime produces the prime-power decomposition of T. Its uniqueness follows from Exercises 38.14–19. Factors of order 1 are omitted, since they contribute only a trivial coordinate.

Exercise 38.21: Recovering the largest torsion coefficient

Because $m_i \mid m_{i+1}$, the valuations $v_p(m_i)$ are nondecreasing for each prime p. The largest exponent for p in the entire prime-power decomposition is consequently $v_p(m_r)$. The same intrinsic largest exponent is $v_p(n_s)$ in the other decomposition. Therefore, writing it as h_p,

$$m_r = \prod_{p \mid\mid T\mid} p^{h_p} = n_s.$$

All primes dividing T occur in this product, so equality of the prime valuations proves equality of the integers.

Exercise 38.22: Recovering all torsion coefficients

For each prime, list its exponents in the unique prime-power decomposition in increasing order. Pad each prime's list on the *left* with zeros to a common length. The last torsion coefficient is the product of the prime powers in the last column; the preceding coefficient is the product in the next-to-last column, and so on. This reconstructs m_{r-1} and n_{s-1} identically: for each prime take its second-largest exponent, or zero if it appears only once. Continue using third-largest exponents and so forth. Divisibility forces precisely this alignment of the lists, because valuations are nondecreasing. The number of columns is the largest number of nonzero exponents in any prime's list: otherwise the first coefficient would be 1, contrary to $m_1, n_1 > 1$. Thus the common length is both r and s, and every coefficient agrees. This proves $r = s$ and $m_i = n_i$ for all i.

39 Free Groups

Exercise 39.1: Reduced words and their inverses

Plan. Combine only *adjacent* powers of the same letter; for an inverse, reverse the syllables and negate every exponent. **(a)** In $a^2b^{-1}b^3a^3c^{-1}c^4b^{-2}$ the adjacent b powers combine to b^2, and the adjacent c powers to c^3. Thus the reduced word and its inverse are

$$a^2b^2a^3c^3b^{-2}, \qquad b^2c^{-3}a^{-3}b^{-2}a^{-2}.$$

(b) In $a^2a^{-3}b^3a^4c^4c^2a^{-1}$, combine the first two a powers and the two c powers. The answers are

$$a^{-1}b^3a^4c^6a^{-1}, \qquad ac^{-6}a^{-4}b^{-3}a.$$

Multiplying either displayed word by its displayed inverse cancels from the middle outward to the empty word, checking the results.

Exercise 39.2: The same products in a free abelian group

Now commuting letters may be regrouped, so add *all* exponents of each letter. **(a)** The exponent totals are $a : 2 + 3 = 5$, $b : -1 + 3 - 2 = 0$, $c : -1 + 4 = 3$. The product is a^5c^3 and its inverse is $a^{-5}c^{-3}$. **(b)** The totals are $a : 2 - 3 + 4 - 1 = 2$, $b : 3$, $c : 4 + 2 = 6$. The product is $a^2b^3c^6$ and its inverse is $a^{-2}b^{-3}c^{-6}$. Unlike Exercise 39.1, rearrangement is allowed because this group is abelian.

Exercise 39.3: Homomorphisms from a rank-two free group

Let its basis be x, y. Any choice of the two images defines one and only one homomorphism, by the universal property of a free group. No commuting condition is imposed. Thus the number of maps to a finite group H is $|H|^2$. The answers are **(a)** $4^2 = 16$, **(b)** $6^2 = 36$, and **(c)** $|S_3|^2 = 36$.

Exercise 39.4: Surjections from a rank-two free group

The images of the two basis elements must generate the codomain, and every ordered generating pair gives exactly one surjection. **(a)** In $\mathbb{Z}_4$, a pair generates if at least one entry is odd. Of 16 pairs, exactly four have both entries even, leaving 12. **(b)** In $\mathbb{Z}_6$, the reductions must generate modulo both 2 and 3. The Chinese remainder theorem makes the choices independent: modulo 2 there are $2^2 - 1 = 3$ nonzero pairs, and modulo 3 there are $3^2 - 1 = 8$. Thus there are $3 \cdot 8 = 24$. **(c)** In S_3, an ordered pair generates exactly when it consists of two distinct transpositions ($3 \cdot 2 = 6$ choices), or one transposition and one 3-cycle (two positions, three transpositions, two cycles: 12 choices). The other pairs lie in a proper cyclic subgroup. Thus there are $6 + 12 = 18$ surjections.

Exercise 39.5: Homomorphisms from a rank-two free abelian group

A map from $\mathbb{Z}^2$ is determined by an ordered *commuting* pair of images. Every pair commutes in the cyclic codomains, so **(a)** and **(b)** again give 16 and 36. For **(c)**, count commuting pairs in S_3. The

identity commutes with all six elements. Each of three transpositions has a centralizer of size 2, and each of two 3-cycles has a centralizer of size 3. Thus the answer is

$$6 + 3 \cdot 2 + 2 \cdot 3 = 18.$$

For any commuting pair (g, h) the homomorphism is $(m, n) \mapsto g^m h^n$, verifying existence as well as the count.

Exercise 39.6: Surjections from a rank-two free abelian group

For the abelian codomains the generating-pair counts are unchanged from Exercise 39.4: **(a)** 12 maps onto $\mathbb{Z}_4$ and **(b)** 24 onto $\mathbb{Z}_6$. **(c)** There are 0 maps onto S_3, since the image of an abelian group is abelian: $f(x)f(y) = f(x + y) = f(y + x) = f(y)f(x)$. A nonabelian group therefore cannot be the entire image.

Exercise 39.7: Definition of a reduced word

The definition is correct. Adjacent syllables involving the same letter can be combined by adding their exponents, and a syllable of exponent zero can be deleted. A word is reduced exactly when neither elementary contraction is possible. The empty word is reduced as well.

Exercise 39.8: Definition of free-group rank

Replace "a set of generators" by "a basis," meaning a free generating set. The rank of a free group is the cardinality of a basis. An arbitrary generating set can contain redundant elements and therefore need not have the invariant size of a basis.

Exercise 39.9: Reflecting on an apparently obvious assertion

Consider the assertion that the final reduced word does not depend on the order of contractions. It is plausible from examples, but examples alone do not prove it. One systematic justification expands syllables into individual letters $a^{\pm 1}$ and scans left to right with a stack: cancel the next letter when it is inverse to the stack's top; otherwise append it. The stack after a prefix is the reduced representative of that prefix. Deleting an adjacent inverse pair anywhere leaves the final stack unchanged, so any sequence of such deletions gives the same result. Combining adjacent equal-letter powers is equivalent to these cancellations together with collecting identical adjacent signs. Thus the apparent obviousness can be replaced by an invariant algorithm. It is reasonable to accept the assertion while learning the main construction, while recognizing what a complete proof needs.

Exercise 39.10: True or false: free groups

(a) True: the subgroup theorem for free groups (Nielsen–Schreier), stated in the section, includes proper subgroups.

(b) False. $2\mathbb{Z} \times \mathbb{Z}$ is a proper subgroup of the free abelian group $\mathbb{Z}^2$, but is abelian of rank 2 and hence is not a free group.

(c) False. $\mathbb{Z} \twoheadrightarrow \mathbb{Z}_2$ is a homomorphism from a free group, but a nontrivial finite group is not free.

(d) True: this is the defining property of free abelian groups.

(e) False. The finitely generated abelian group $\mathbb{Z}_2$ is not free abelian, because it has nonzero torsion.

(f) False. A free group is, by definition, free on a basis; $\mathbb{Z}$ is a basic example.

(g) False. $\mathbb{Z}$ is both free abelian and free as a group on one generator.

(h) True. A free group on at least two basis elements is nonabelian (the reduced words xy and yx differ), whereas a free abelian group is abelian. A rank-greater-than-one free abelian group is neither trivial nor cyclic, so it cannot instead be a free group of rank 0 or 1.

(i) False. $\mathbb{Z}$ and the free group on two letters have different abelian/nonabelian behavior.

(j) True. A bijection between bases extends to an isomorphism by matching finite integer combinations.

Exercise 39.11: A basis allowing torsion

(a) In $\mathbb{Z}_4$ the relation $1 \cdot 2 + 2 \cdot 3 = 0$ holds, but the individual summands are both 2, not zero. Thus $\{2, 3\}$ fails this exercise's independence condition. The singleton $\{1\}$ is a basis: it generates, and a one-term relation already has its one term zero. **(b)** The same singleton $\{1\}$ is a basis of $\mathbb{Z}_6$. Also $\{2, 3\}$ generates because $3 - 2 = 1$. If $2m + 3n = 0 \pmod 6$, reducing modulo 3 gives $m = 0 \pmod 3$, and reducing modulo 2 gives $n = 0 \pmod 2$. Consequently $2m = 0$ and $3n = 0$ individually. This proves the required independence. **(c)** Yes. A Section 38 basis forces each integer coefficient to be zero in a zero sum, which in particular forces each summand to be zero. It also generates the group. **(d)** Write a finite abelian group as $\mathbb{Z}_{d_1} \times \cdots \times \mathbb{Z}_{d_n}$ with $d_i \mid d_{i+1}$, using its invariant-factor decomposition. Let b_i be the inverse image of the standard generator in coordinate i. They generate, have orders d_i, and a zero sum has zero in each coordinate, forcing each summand $m_i b_i$ to be zero. Thus they form a basis in the present torsion-allowing sense.

Exercise 39.12: The universal abelian quotient

(a) If $\phi_i : G \twoheadrightarrow G_i^*$ have the stated property, there are maps $\theta_1 : G_2^* \to G_1^*$ and $\theta_2 : G_1^* \to G_2^*$ with $\phi_1 = \theta_1\phi_2$ and $\phi_2 = \theta_2\phi_1$. Then $(\theta_1\theta_2)\phi_1 = \phi_1$. Since ϕ_1 is onto, $\theta_1\theta_2$ is the identity on G_1^*. Similarly $\theta_2\theta_1$ is the identity on G_2^*, so the groups are isomorphic. **(b)** Let $G^* = G/[G, G]$ and let ϕ be the quotient map. For a homomorphism $\psi : G \to A$ with A abelian, every commutator maps to the identity, so $[G, G] \subseteq \ker \psi$. Therefore $\theta(g[G, G]) = \psi(g)$ is well defined, is a homomorphism, and satisfies $\psi = \theta\phi$. **(c)** This is the abelianization of G, its largest universal abelian quotient. The factor map is unique because ϕ is onto.

Exercise 39.13: The universal free group

(a) Let $u : G_1 \to G_2$ and $v : G_2 \to G_1$ be the unique homomorphisms with $ug_1 = g_2$ and $vg_2 = g_1$. Then $vug_1 = g_1$, and both vu and the identity are homomorphisms extending g_1. Uniqueness forces $vu = \mathrm{id}_{G_1}$. Similarly $uv = \mathrm{id}_{G_2}$, giving an isomorphism. For clarity the universal property also implies that g_i is injective: distinguish two points of S by a function to $\mathbb{Z}_2$ and factor it. It implies that $g_i(S)$

generates: for $H = \langle g_i(S) \rangle$, factor $S \to H$ through a homomorphism $G_i \to H$; its composite with the inclusion into G_i is the identity by uniqueness, hence $H = G_i$. **(b)** Construct $F[S]$ as the group of reduced words on the alphabet S, with $g(s)$ the one-letter word. Any function $f : S \to G'$ extends by replacing letters with their images and using integer powers. Cancellation preserves the result; this is the unique extending homomorphism. **(c)** The "blop group" is exactly the free group on S.

Exercise 39.14: A universal property for free abelian groups

For a set S, require an abelian group A and a map $i : S \to A$ such that for every abelian group B and every function $f : S \to B$, there is a unique homomorphism $\overline{f} : A \to B$ with $\overline{f}\, i = f$. Existence follows by taking $A = \bigoplus_{s \in S} \mathbb{Z}$, the finitely supported integer functions on S, and $i(s)$ the function that is 1 at s and zero elsewhere. Define

$$\overline{f}\left(\sum_s n_s i(s)\right) = \sum_s n_s f(s).$$

The sums are finite, and commutativity in B makes this an additive map. Every element has a unique such expansion, so the extension exists and is unique. Conversely the universal property characterizes this group up to a unique isomorphism carrying the specified generators to one another, by the two-composites argument of Exercise 39.13(a). Thus it characterizes the free abelian group with basis S.

40 Group Presentations

Exercise 40.1: Presentations of $\mathbb{Z}_4$

One, two, and three generators can be used as follows:

$$\langle a \mid a^4 = 1\rangle, \qquad \langle a, b \mid a^4 = 1,\ b = a^2\rangle,$$

$$\langle a, b, c \mid a^4 = 1,\ b = a^2,\ c = a^3\rangle.$$

In each case eliminate the additional generators using their defining relations. Every element is a power of a, with exponent reduced modulo 4, so there are at most four elements. Sending a to $1 \in \mathbb{Z}_4$, b to 2 and c to 3 (where present) satisfies the relations and maps onto $\mathbb{Z}_4$, proving there are exactly four elements.

Exercise 40.2: A three-generator presentation of S_3

Use

$$\langle a, b, c \mid a^3 = 1,\ b^2 = 1,\ bab = a^{-1},\ c = ab\rangle.$$

The relation for c lets us eliminate it. The conjugation relation gives $ba = a^{-1}b$, so every word becomes $a^i b^j$ with $0 \le i < 3$, $0 \le j < 2$. Thus there are at most six elements. Map $a \mapsto (123)$, $b \mapsto (12)$, $c \mapsto (123)(12)$. These permutations satisfy the relations and generate S_3, so the presented group has at least six elements. It is therefore isomorphic to S_3.

Exercise 40.3: Tables for the octic and quaternion groups

Plan. Write every element as $a^i b^j$, with $0 \le i < 4$, $j \in \{0, 1\}$. The rule $ba = a^{-1}b$ moves b past a power of a. If $b^2 = a^c$, then

$$(a^i b^j)(a^u b^v) = a^{\,i+(-1)^j u + cjv} b^{\,j+v \bmod 2},$$

where the exponent of a is reduced modulo 4. Use $c = 0$ for the octic group and $c = 2$ for the quaternion group. The octic table, in the requested order, is

$\cdot$	1	a	a^2	a^3	b	ab	a^2b	a^3b
1	1	a	a^2	a^3	b	ab	a^2b	a^3b
a	a	a^2	a^3	1	ab	a^2b	a^3b	b
a^2	a^2	a^3	1	a	a^2b	a^3b	b	ab
a^3	a^3	1	a	a^2	a^3b	b	ab	a^2b
b	b	a^3b	a^2b	ab	1	a^3	a^2	a
ab	ab	b	a^3b	a^2b	a	1	a^3	a^2
a^2b	a^2b	ab	b	a^3b	a^2	a	1	a^3
a^3b	a^3b	a^2b	ab	b	a^3	a^2	a	1

The quaternion table is

$\cdot$	1	a	a^2	a^3	b	ab	a^2b	a^3b
1	1	a	a^2	a^3	b	ab	a^2b	a^3b
a	a	a^2	a^3	1	ab	a^2b	a^3b	b
a^2	a^2	a^3	1	a	a^2b	a^3b	b	ab
a^3	a^3	1	a	a^2	a^3b	b	ab	a^2b
b	b	a^3b	a^2b	ab	a^2	a	1	a^3
ab	ab	b	a^3b	a^2b	a^3	a^2	a	1
a^2b	a^2b	ab	b	a^3b	1	a^3	a^2	a
a^3b	a^3b	a^2b	ab	b	a	1	a^3	a^2

For example, the product $(a^2b)(a^3b)$ is $a^{2-3} = a^3$ in the first group and $a^{2-3+2} = a$ in the second. This checks the effect of the different square of b.

Exercise 40.4: All groups of order 14

Sylow gives a unique order-7 subgroup $\langle a\rangle$, so it is normal. Cauchy's theorem gives b of order 2 outside it. The elements a^ib^j exhaust the group, since the subgroup generated by a, b has order divisible by both 7 and 2. Conjugation has $bab^{-1} = a^r$ with r nonzero modulo 7. Since $b^2 = 1$, $r^2 \equiv 1 \pmod 7$, so $(r-1)(r+1) = 0$ in the field $\mathbb{Z}_7$ and $r = 1$ or 6. The first case gives $\mathbb{Z}_{14}$. The second gives $\langle a, b \mid a^7 = b^2 = 1,\ ba = a^6b\rangle$, the dihedral group D_7. Exercise 40.13 proves it has order 14; it is nonabelian because inversion acts nontrivially on a. These are the only two possibilities, and their different commutativity makes them nonisomorphic.

Exercise 40.5: All groups of order 21

The order-7 Sylow subgroup $\langle a\rangle$ is normal. Choose b of order 3 by Cauchy's theorem; it is outside $\langle a\rangle$, and together the elements generate all 21 elements. The action $bab^{-1} = a^r$ must satisfy $r^3 \equiv 1 \pmod 7$. Checking the six nonzero residues gives $r = 1, 2, 4$. For $r = 1$ the group is $\mathbb{Z}_{21}$. For $r = 2$ we obtain

$$\langle a, b \mid a^7 = b^3 = 1,\ ba = a^2b\rangle,$$

a nonabelian group of order 21 by Exercise 40.13. The choice $r = 4$ gives the same isomorphism type: replacing b by b^2 changes the action exponent from 2 to $2^2 = 4$, while b^2 still generates the order-3 subgroup. Thus there is exactly one abelian and one nonabelian isomorphism type.

Exercise 40.6: Consequences of relators

Include conjugates, which are missing from the proposed definition. A consequence is an element of the normal closure of the relators in the free group, equivalently a finite product

$$\prod_{j=1}^{k} w_j r_{i_j}^{\epsilon_j} w_j^{-1}, \qquad \epsilon_j \in \{1, -1\}.$$

Allowing arbitrary integer powers gives the same normal closure. The empty product is the identity. Products merely of powers of the original relators need not form a normal subgroup.

Exercise 40.7: Isomorphic presentations

Two presentations are isomorphic when the groups they present are isomorphic. A bijective renaming of generators and relators is sufficient but not necessary. For example $\langle a \mid a^4 = 1\rangle$ and $\langle a, b \mid a^4 = 1, b = a^2\rangle$ both present $\mathbb{Z}_4$, as proved in Exercise 40.1, but their generator sets have different sizes, so no bijective renaming exists. The printed proposed definition is therefore too restrictive.

Exercise 40.8: True or false: presentations

(a) True. Take a free group mapping onto the group and use all kernel elements as relators.

(b) True. Introduce arbitrarily many new generators together with relations expressing each as an old word.

(c) False. By the definition used here, all presentations of the same group are isomorphic presentations.

(d) False. A finite presentation gives a finitely generated, hence countable, group. The uncountable additive group $\mathbb{R}$ cannot have one.

(e) False. $\langle a \mid \rangle$ presents the infinite cyclic group with a finite generator set and no relators.

(f) True. A cyclic group has $\langle a \mid a^n = 1\rangle$ if its order is n, or $\langle a \mid \rangle$ if it is infinite. For the trivial group use $n = 1$.

(g) True. Consequences form the normal closure, which contains every conjugate of each relator.

(h) False. $\langle a \mid a^2 = 1\rangle$ and $\langle a \mid a^3 = 1\rangle$ have the same number of generators but give nonisomorphic groups of orders 2 and 3.

(i) True. If F/R is abelian, all commutators in F map to the identity, so $[F, F] \subseteq R$.

(j) False. $\langle a, b \mid b = 1\rangle$ presents the free group $\mathbb{Z}$ but has the nonidentity free word b as a relator.

Exercise 40.9: Groups of order 15 are abelian

Sylow gives a normal subgroup $\langle a\rangle$ of order 5. Cauchy's theorem gives b of order 3, and a, b generate the group. If $bab^{-1} = a^r$, conjugating three times gives $r^3 \equiv 1 \pmod 5$. Among the nonzero residues modulo 5 only $r = 1$ satisfies this (or use that their multiplicative group has order 4, coprime to 3). Thus a and b commute. The group is generated by commuting elements, hence abelian; indeed ab has order 15, so it is cyclic. There is no nonabelian group of order 15.

Exercise 40.10: An order-six nonabelian presentation

Apply Exercise 40.13 with $m = 3, n = 2, r = 2$. Since $2^2 \equiv 1 \pmod 3$, its normal-form model has exactly $3 \cdot 2 = 6$ elements. It realizes a with order 3 and $ba = a^2b$. If a, b commuted, cancellation would give $a = a^2$, hence $a = 1$, contradicting its order 3. Thus the group is nonabelian.

Exercise 40.11: Uniqueness of the nonabelian group of order six

For any group of order 6, the Sylow 3-subgroup $\langle a\rangle$ is unique and normal. An element b of order 2 exists by Cauchy's theorem and lies outside it, so a, b generate. Conjugation sends a to a or a^2, the two generators of $\langle a\rangle$. The first case makes the group abelian and cyclic of order 6. The second gives exactly the presentation in Exercise 40.10, which has order 6 and thus maps isomorphically onto the given group. Since S_3 is nonabelian of order 6, it realizes the second case. Concretely one may send a to (123) and b to (12).

Exercise 40.12: No subgroup of order six in A_4

An order-six subgroup would, by Exercise 40.11, be cyclic of order 6 or isomorphic to S_3. The cyclic possibility is impossible because elements of A_4 have orders only $1, 2, 3$: they are the identity, double transpositions, and 3-cycles. The S_3 possibility is also impossible. Its three involutions do not commute, and the product of two distinct involutions has order 3. In A_4, the only involutions are the three double transpositions; they form a Klein four group, commute, and multiply in pairs to another involution. An isomorphism preserves these products and orders, giving a contradiction.

Exercise 40.13: The normal-form multiplication criterion

(a) Write a formal element as (s, t) with $s \in \mathbb{Z}_m$ and $0 \le t < n$, and define

$$(s,t)(u,v) = (s + r^t u \bmod m,\ t + v \bmod n).$$

Assume $r^n \equiv 1 \pmod m$. Then r is invertible modulo m, and replacing $t+v$ by its residue modulo n leaves r^{t+v} unchanged modulo m. Thus the first coordinates of both associations of a triple product equal $s + r^t u + r^{t+v} w$, and the second equal $t + v + z \pmod n$. The operation is associative. The identity is $(0, 0)$. For (s, t) take $v = -t \pmod n$ and $u = -r^{-t} s \pmod m$; the resulting (u, v) is a two-sided inverse. This proves a group of order mn. Conversely, for $n > 1$ suppose the operation is a group. Then $a = (1, 0)$ has order m, $b = (0, 1)$ has order n, and $bab^{-1} = a^r$. Repeating conjugation n times gives $a = a^{r^n}$, hence $r^n \equiv 1 \pmod m$. *Degenerate case:* if $n = 1$, the displayed operation is always just addition in $\mathbb{Z}_m$, independent of r. Thus the literal "if and only if" in part (a) needs $n > 1$; its sufficiency remains valid for $n = 1$. **(b)** If the congruence holds, the group constructed above satisfies the presentation relations. Conversely those relations put every word in one of the mn forms $a^i b^j$, so the presented group maps onto this order-mn model and has at most mn elements; it therefore has exactly mn. If the presentation has order mn, its mn normal forms must all be distinct. In particular a has order m. The relations $b^n = 1$ and $bab^{-1} = a^r$ imply $a = a^{r^n}$ and hence the congruence. This also proves part (b) for $n = 1$, where $b = 1$ forces $a = a^r$.

Exercise 40.14: A unique nonabelian group of order pq

Let $p < q$ be primes with $p \mid q - 1$. For any order-pq group, the Sylow q-subgroup is unique because its number divides $p < q$ and is 1 modulo q. Write it $\langle a\rangle$, and choose b of order p by Cauchy's theorem. The two subgroups intersect trivially and generate the group, so it has a presentation

$$\langle a, b \mid a^q = b^p = 1,\ ba = a^r b\rangle, \qquad r^p \equiv 1 \pmod q.$$

The unit group $\mathbb{Z}_q^*$ is cyclic of order $q - 1$, so the pth roots of 1 form its unique order-p subgroup. Its nonidentity elements are $r, r^2, \ldots, r^{p-1}$ for any one generator r. The root 1 gives the abelian

cyclic group. Every other root gives a nonabelian order-pq group by Exercise 40.13. Replacing b by b^j, $1 \leq j < p$, changes the action exponent to r^j without changing its generated order-p subgroup. Therefore all nontrivial roots give isomorphic presentations, proving existence and uniqueness of the nonabelian isomorphism type.